Student Number: ____________________

2022
Higher School Certificate
Trial Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black/blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For Section II, all relevant mathematical reasoning and calculations must be shown
- Write your student number and/or name at the top of every page

Total marks – 100

Section I – Pages 3 – 8
10 marks
- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – Pages 10 – 34
90 marks
- Attempt Questions 11 – 32
- Allow about 2 hour and 45 minutes for this section

BLANK PAGE

2022
Higher School Certificate
Trial Examination

Mathematics Advanced

Booklet 1 (40 marks)

Student Number: ____________________________

Section I (10 marks)

Attempts Questions 1 – 10

Allow about 15 minutes for this section.

Use the multiple-choice grid on page 7 for questions 1 – 10.

---

1 What is the domain of the function $y = \sqrt{2x-1}$?

(A) $\left(\frac{1}{2}, \infty\right)$

(B) $\left[\frac{1}{2}, \infty\right)$

(C) $\left(\frac{1}{2}, \infty\right]$

(D) $\left[\frac{1}{2}, \infty\right]$

2 Which of the following is equivalent to $sec^2x - sin^2x - cos^2x$?

(A) $tan^2x + 1$

(B) $1 - tan^2x$

(C) $tan^2x - 1$

(D) tan^2x

3 Which interval gives the range of the function $y = -(1 + 3sin2x)$?

(A) $-4 \leq y \leq -2$

(B) $-5 \leq y \leq -1$

(C) $-1 \leq y \leq 5$

(D) $-4 \leq y \leq 2$

4 What transformation is required to produce the graph of $y = 3^{x+1}$ from the graph of $y = 3^x$?

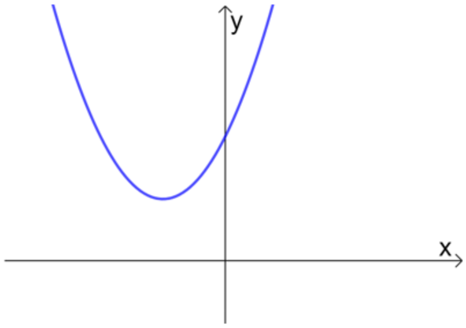
(A) Shift left by 3 units

(B) Shift right by 1 unit

(C) Dilate vertically by a factor of 3

(D) Dilate horizontally by a factor of 3

5 The diagram shows the graph of $y = x^2 + bx + 2$.



Which of the following could be the value of b?

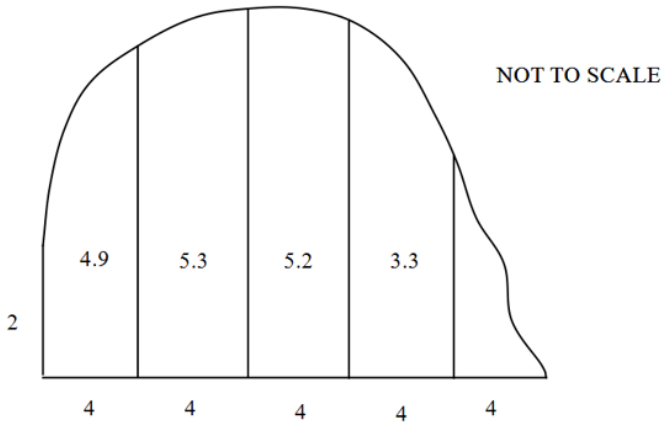
(A) 2

(B) -2

(C) 3

(D) -3

6 The diagram below shows a native garden. All measurements are in metres.



What is an approximate value for the area of the native garden using the trapezoidal rule with 5 intervals?

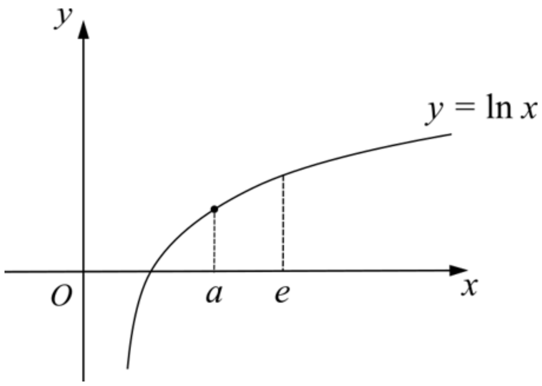
(A) $39\,m^2$

(B) $58\,m^2$

(C) $72\,m^2$

(D) $79\,m^2$

7 The line $y = mx + c$ is a tangent to the curve $y = lnx$ at the point where $x = a$, as shown in the diagram.



Which of the following statements is true?

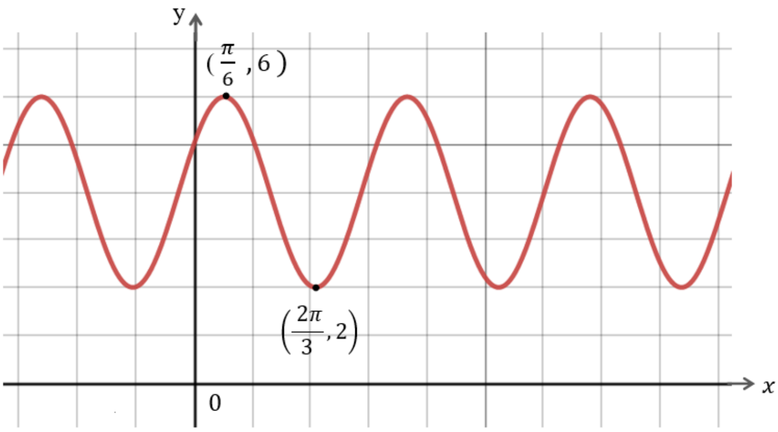
(A) $\frac{1}{e} < m < 1$ and $-1 < c < 0$

(B) $1 < m < e$ and $-1 < c < 0$

(C) $\frac{1}{e} < m < 1$ and $0 < c < 1$

(D) $1 < m < e$ and $0 < c < 1$

8 Which of the equations below is represented by the graph?



(A) $y = 4 + 2\cos\left(2x - \frac{\pi}{4}\right)$

(B) $y = 4 + 2\cos\left(2x - \frac{\pi}{3}\right)$

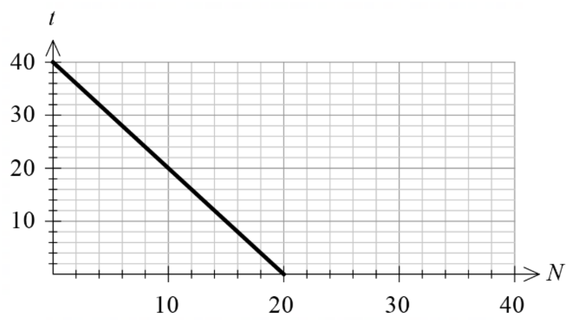
(C) $y = 4 + 2\cos\left(2x + \frac{\pi}{6}\right)$

(D) $y = 4 + 2\cos\left(2x - \frac{\pi}{6}\right)$

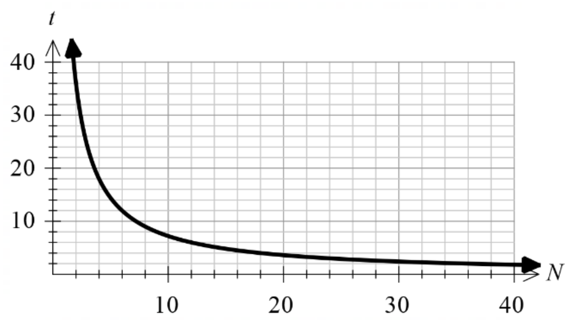
9 The time taken (t days) to paint a house varies inversely with the number of painters (N) on the job.

Which of the following best represents the relationship between the number of painters and the time taken to paint a house, given that it takes 4 painters 18 days to paint the same house?

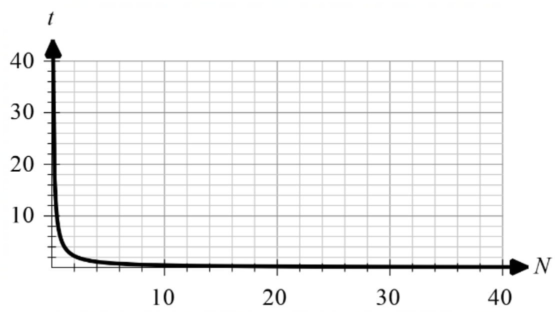
(A)



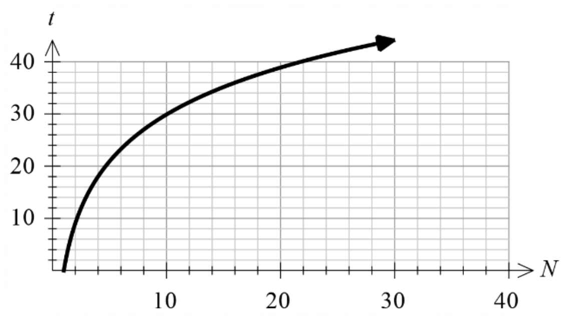
(B)



(C)



(D)



10 The functions $F(x), G(x)$ and $H(x)$ are defined for all real x and the function $H(x) = F\left(\frac{1}{G\,(x)}\right)$. Given that $G(3) = \frac{1}{2}$, $G'(3) = \frac{1}{10}$ and $F'(2) = 5$, what is the gradient of the tangent to the curve $H(x)$ at $x = 3$?

(A) 20

(B) -20

(C) 2

(D) -2

End of Section I

Record your answers below by shading the appropriate box.

1	**A**	**B**	**C**	**D**
2	**A**	**B**	**C**	**D**
3	**A**	**B**	**C**	**D**
4	**A**	**B**	**C**	**D**
5	**A**	**B**	**C**	**D**
6	**A**	**B**	**C**	**D**
7	**A**	**B**	**C**	**D**
8	**A**	**B**	**C**	**D**
9	**A**	**B**	**C**	**D**
10	**A**	**B**	**C**	**D**

2022
Higher School Certificate
Trial Examination

Mathematics Advanced
Section II

90 marks

Attempt Questions 11 – 32

Allow about 2 hours and 45 minutes for this section

Instructions

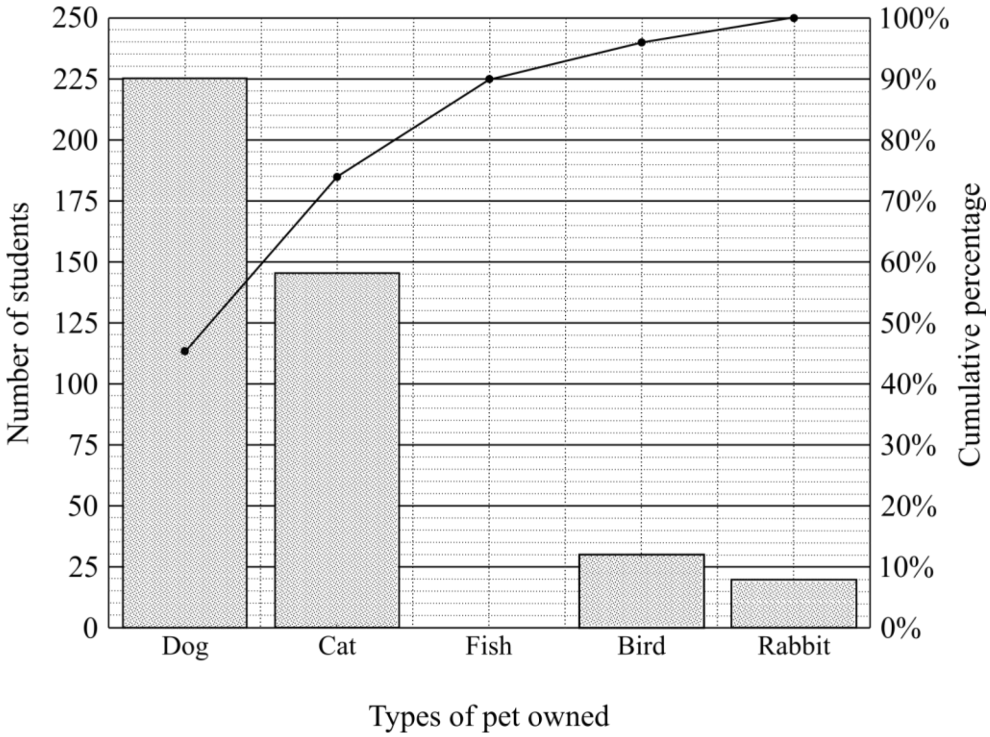
- Answer each question in the space provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

MARKS

Question 11 (4 marks)

A group of students was surveyed and data relating to the types of pets they owned was collected. The Pareto chart shows the data collected. The column representing the number of students owning pet fish has been removed.



(a) How many students own a pet dog or cat? **1**

....

....

(b) Complete the Pareto chart, showing the number of students who own a pet fish. Show all necessary working out in the space below. **3**

....

....

....

....

Question 12 (3 marks)

A table for the present value for an annuity of $1 is shown.

The present value of an annuity of $1

Period	1%	2%	4%	6%	8%	10%
1	0.9901	0.9804	0.9615	0.9434	0.9259	0.9091
2	1.9704	1.9416	1.8861	1.8334	1.7833	1.7355
3	2.941	2.8839	2.7751	2.673	2.5771	2.4869
4	3.902	3.8077	3.6299	3.4651	3.3121	3.1699
5	4.8534	4.7135	4.4518	4.2124	3.9927	3.7908
6	5.7955	5.6014	5.2421	4.9173	4.6229	4.3553

(a) What would be the present value of a $5000 per year annuity at 6% per annum for six years, with interest compounding annually? **1**

....

....

....

(b) An annuity of $2000 per quarter is invested at 8% per annum, compounded quarterly for one year. What is the present value of the annuity? **1**

....

....

....

(c) An annuity has a present value of $46 000. **1**

What is the amount of each yearly payment, compounded at 8% per annum, that is required to accumulate this annuity after five years?

....

....

....

Question 13 (3 marks)

Solve $2cos^2x = -3sinx$ for $0 \leq x \leq 2\pi$. **3**

....

....

....

....

....

....

Question 14 (2 marks)

Evaluate $\int_0^{\frac{\pi}{2}} \cos 2x \ dx$ **2**

....

....

....

....

Question 15 (2 marks)

A and B are two mutually exclusive events. **2**

If $P(B) = 2P(A)$ and P (A ∪ B) = $\frac{4}{9}$, what is the value of P (A) ?

....

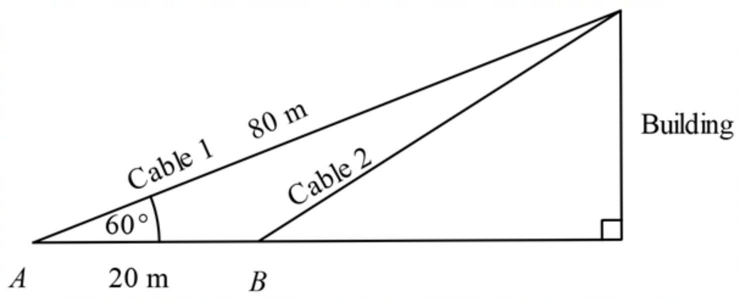
....

....

....

Question 16 (4 marks)

A support cable, 80 metres long, has been secured at an angle of elevation of 60° from point *A* to the top of a building. A second support cable will be secured from point *B*, 20 metres closer to the base of the building than point *A*. This information is shown in the diagram.



(a) Show that the length of the second cable is approximately 72 metres. **2**

....

....

....

....

(b) Given that the height of the building is 69.2 metres, calculate the distance between point *A* and the base of the building. **2**

Give your answer to the nearest metre.

....

....

....

....

Question 17 (2 marks)

Find $\int \frac{x}{\sqrt{9-x^2}}\ dx$. **2**

....

....

....

....

Question 18 (4 marks)

(a) Show that $\frac{d}{dx}(3x^2e^{2x+1}) = 6xe^{2x+1}\ (x+1)$. **2**

....

....

....

....

(b) Hence evaluate $\int_1^2 x(x+1)e^{2x+1}\ dx$. **2**

....

....

....

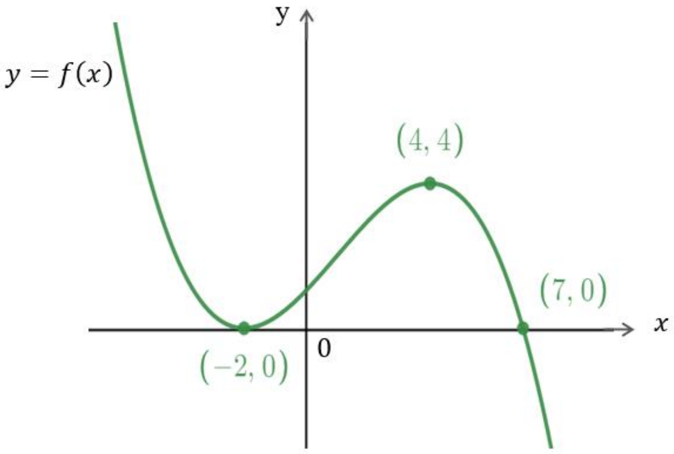
....

....

....

Question 19 (3 marks)

The diagram shows the graph of the function $y = f(x)$. **3**



In the space below, sketch the graph of the function $y = -2f\left(\frac{x}{2}\right)$ showing the coordinates of the turning points and the x intercepts.

....

....

....

....

....

....

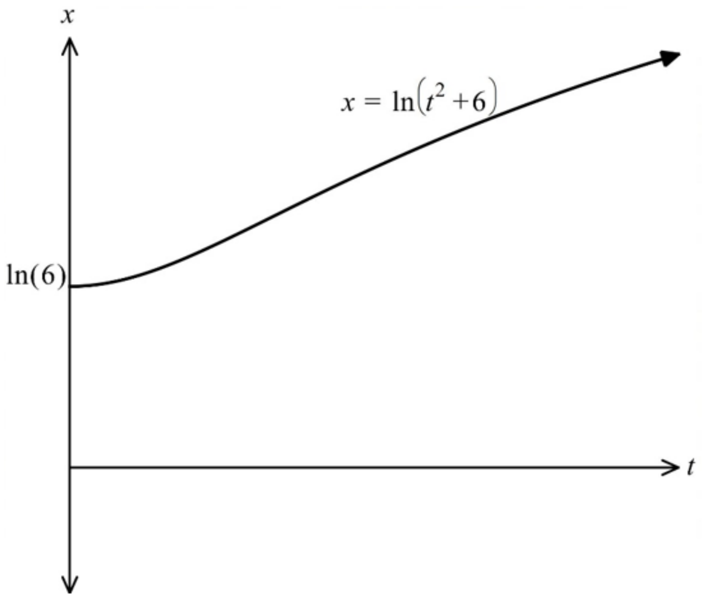
....

....

Booklet I continues on the next page

Question 20 (3 marks)

A particle moves on the x-axis so that its displacement in metres from the origin at a time t seconds is given by the equation $x = \ln(t^2 + 6)$. The particle starts from rest at the point $x = \ln(6)$ and accelerates in a positive direction as shown in the distance-time graph below. **3**



Determine when the acceleration of the particle becomes zero and find the velocity at this time.

....

....

....

....

....

....

....

End of Booklet 1

This space can be used if needed. Ensure that you label any answers written here clearly giving their question number and part letter.

2022
Higher School Certificate
Trial Examination

Mathematics Advanced

Booklet 2 (30 marks)

Student Number: ______________________________

MARKS

Question 21 (4 marks)

Find the equation of the normal to the curve $y = x^4 - 3x^2 + 18x + 24$ at the point where $x = -2$. **4**

....

....

....

....

....

....

....

Question 22 (4 marks)

Find the greatest and least values of the function $f(x) = 10x^2e^{-x}$ in the domain [0, 4]. **4**

....

....

....

....

....

....

....

....

Question 23 (3 marks)

A continuous random variable X has a probability density function f given by: **3**

$$f(x) = \begin{cases} Ax + B & 1 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases}$$

where *A* and *B* are constants. The median is 3. Find the values of *A* and *B*.

....

....

....

....

....

....

....

....

Question 24 (4 marks)

At time t hours after midnight on a particular day, the depth D metres of water in the Oztown harbour is given by the function:

$$D = 8 + 2\cos\left(\frac{4\pi}{25}t\right), \text{ for } 0 \leq t \leq 24$$

(a) Find the amplitude and period of the function. **2**

....

....

....

....

Question 24 continues on the next page

(b) Sketch the graph of D as a function of t, clearly showing the times on that day when the water level was at its lowest and highest. **2**

Question 25 (4 marks)

The height of a tree in a park is modelled by $h = 750 - 650\ a^{-0.5t}$, where h is the height of the tree in centimetres (cm) and t is its age in years since the tree was planted in the park. The tree is 490 cm high after 2 years. **4**

Find the value of t, correct to one decimal place, for which the instantaneous rate of increase is 12 cm per year.

....

....

....

....

....

....

....

....

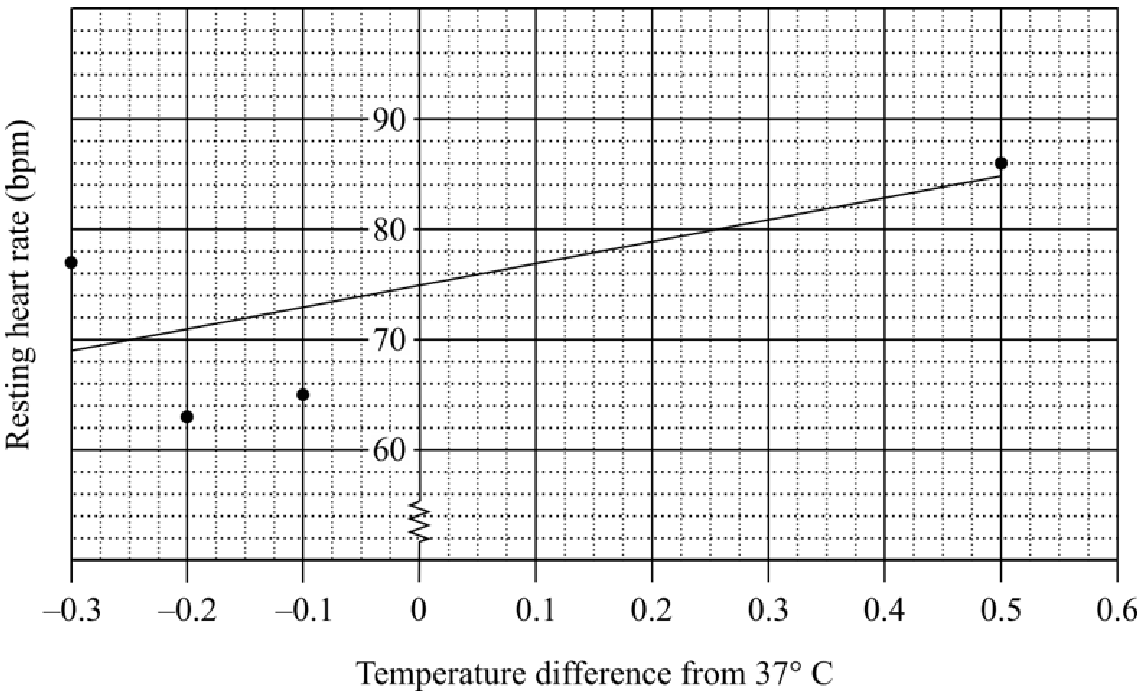
Question 26 (6 marks)

A healthy human body temperature is $37.0°C$.

Eight randomly selected people were examined by medical staff. The difference in their body temperature from $37.0°C$ (in degrees) and resting heart rate (in beats per minute) were recorded.

Temperature difference from 37ºC (x)	–0.2	–0.3	–0.3	–0.2	–0.1	0	0.2	0.5
Heart rate (y)	63	77	70	74	65	78	79	86

(a) Complete the scatterplot by adding the last four points from the table. **1**



(b) The least-squares regression line has been plotted in the graph in part (a). **2**

Find the equation of this line.

....

....

....

....

(c) By using the equation of the regression line, predict the resting heart rate of a person with a body temperature of 37.3° C. Give your answer correct to the nearest whole number. **2**

………

………

………

………

(d) Explain why the least-squares regression line would not be reliable to predict the resting heart rate of a person with a body temperature of 37.6° C? **1**

………

………

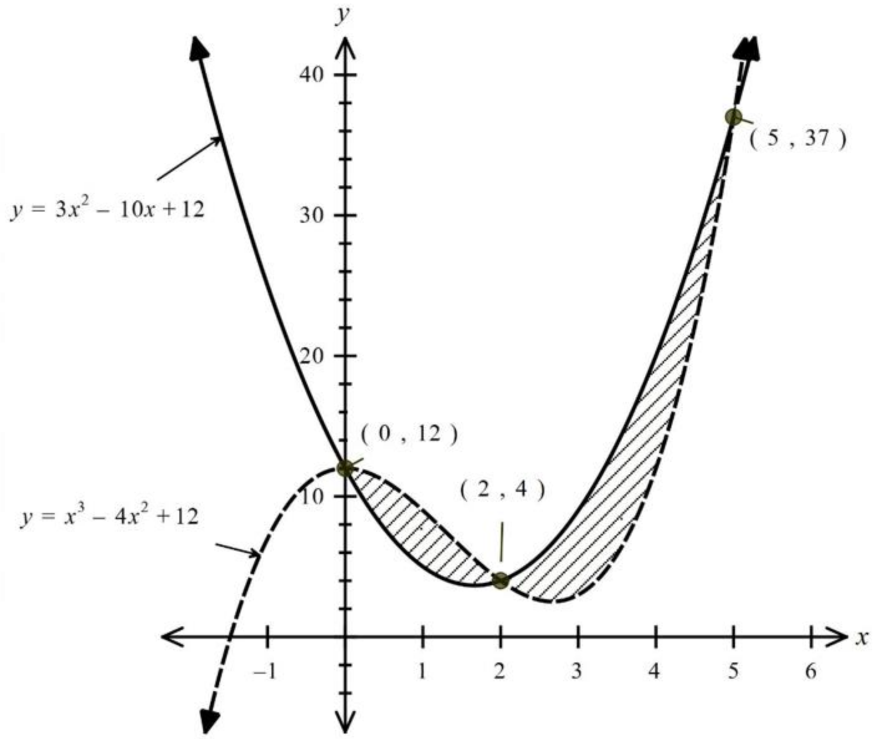
………

………

Booklet 2 continues on the next page

Question 27 (5 marks)

The curves $y = x^3 - 4x^2 + 12$ and $y = 3x^2 - 10x + 12$ intersect at the points (0, 12), (2, 4) and (5, 37) as shown on the graph below. **5**



Calculate the total shaded area enclosed between the two curves.

....

....

....

....

....

....

....

....

End of Booklet 2

This space can be used if needed. Ensure that you label any answers written here clearly giving their question number and part letter.

2022
Higher School Certificate
Trial Examination

Mathematics Advanced

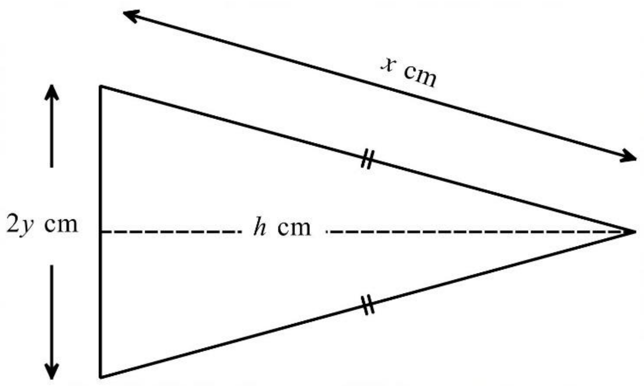
Booklet 3 (30 marks)

Student Number: ____________________________

MARKS

Question 28 (7 marks)

A banner is designed as an isosceles triangle, with equal sides of length x cm and base of length $2y$ cm, as shown.



The total perimeter of the triangle is 40 cm.

(a) Show that the area of the triangle in terms of x can written as: **2**

$$A = (20 - x)\,(40x - 400)^{\frac{1}{2}}$$

....

....

....

....

....

....

....

....

Question 28 continues on the next page

(b) Use calculus to find the values of x and y which give a maximum area and find this area. Give your answer to 2 significant figures. **5**

....

....

....

....

....

....

....

....

....

....

....

....

....

....

Question 29 (4 marks)

Find the coordinates and nature of the stationary points on the curve $y = \frac{x^2}{x-2}$. **4**

....

....

....

....

....

....

....

....

....

....

....

Booklet 3 continues on the next page

Question 30 (6 marks)

Bree borrows $20 000 to buy a car. She is charged interest at 4.8% pa, calculated at the end of each month, and she repays regular monthly instalments of $$P$ after the interest is added.

(a) Show that the amount owing after n months is given by the equation: **2**

$A_n = 20000 \times 1.004^n - P(1.004^{n-1} + 1.004^{n-2} + \cdots \ldots \ + 1.004^2 + 1.004 + 1)$.

....

....

....

....

....

(b) After paying the loan for 12 months, she still owes $17 055.80 on the loan. **2**

Find the amount of the monthly instalment, $$P$.

....

....

....

....

....

....

(c) Determine the number of months required to pay off the loan completely (the term of the loan). **2**

....

....

....

Question 31 (7 marks)

A continuous random variable X has probability density function $f(x)$ given by:

$f(x) = \frac{2}{x^2}$ for $1 \leq x \leq 2$, and $f(x) = 0$ for $x < 1$ or $x > 2$.

(a) Show that $f(x)$ is a probability density function. **2**

....

....

....

....

(b) Find the cumulative distribution function $F(x)$ of the random variable X. **2**

....

....

....

....

(c) Show that if Q_1, Q_2, Q_3 are the quartiles of the distribution X, then $\frac{1}{Q_1}, \frac{1}{Q_2}, \frac{1}{Q_3}$ are consecutive terms in an arithmetic sequence. **3**

....

....

....

....

....

....

Question 32 (6 marks)

A bag contains only strawberry and honey lollies. There are at least 4 lollies of either of the two flavours and the total number of lollies in the bag is n.

Three lollies are to be selected at random from the bag without replacement.

Let X be the number of strawberry lollies selected. The probability distribution table for the number X of strawberry lollies selected is as shown.

x	0	1	2	3
P (X)	$\frac{b}{9}$	b	$log_9\, a$	$\frac{5b}{9}$

(a) Show that $0 \leq b \leq \frac{3}{5}$ **2**

....

....

....

....

....

(b) Given that $E\ (X) = \frac{9}{5}$, find the values of a and b. **3**

....

....

....

....

....

....

Question 32 continues on the next page

(c) Given that the total number of lollies in the bag is $n = 10$, use guess and check to find the number of strawberry lollies in the bag. **1**

....

....

....

....

End of Booklet 3

End of Examination

This space can be used if needed. Ensure that you label any answers written here clearly giving their question number and part letter.

NSW GOVERNMENT

NSW Education Standards Authority

2021 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

and $\alpha\beta\gamma = -\frac{d}{a}$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, \ r \neq 1$$

$$S = \frac{a}{1 - r}, \ |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

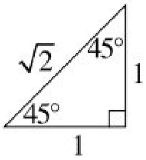
$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$\sin A = \dfrac{\text{opp}}{\text{hyp}}, \quad \cos A = \dfrac{\text{adj}}{\text{hyp}}, \quad \tan A = \dfrac{\text{opp}}{\text{adj}}$

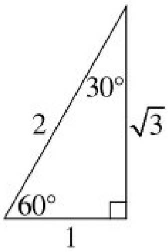
$A = \dfrac{1}{2}ab\sin C$



$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

$c^2 = a^2 + b^2 - 2ab\cos C$

$\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$



$l = r\theta$

$A = \dfrac{1}{2}r^2\theta$

Trigonometric identities

$\sec A = \dfrac{1}{\cos A}, \ \cos A \neq 0$

$\operatorname{cosec} A = \dfrac{1}{\sin A}, \ \sin A \neq 0$

$\cot A = \dfrac{\cos A}{\sin A}, \ \sin A \neq 0$

$\cos^2 x + \sin^2 x = 1$

Compound angles

$\sin(A + B) = \sin A\cos B + \cos A\sin B$

$\cos(A + B) = \cos A\cos B - \sin A\sin B$

$\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A\tan B}$

If $t = \tan\dfrac{A}{2}$ then $\sin A = \dfrac{2t}{1 + t^2}$

$\cos A = \dfrac{1 - t^2}{1 + t^2}$

$\tan A = \dfrac{2t}{1 - t^2}$

$\cos A\cos B = \dfrac{1}{2}\left[\cos(A - B) + \cos(A + B)\right]$

$\sin A\sin B = \dfrac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$

$\sin A\cos B = \dfrac{1}{2}\left[\sin(A + B) + \sin(A - B)\right]$

$\cos A\sin B = \dfrac{1}{2}\left[\sin(A + B) - \sin(A - B)\right]$

$\sin^2 nx = \dfrac{1}{2}(1 - \cos 2nx)$

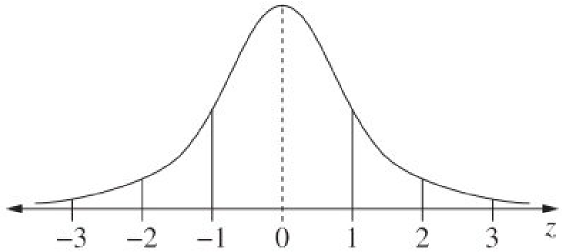
$\cos^2 nx = \dfrac{1}{2}(1 + \cos 2nx)$

Statistical Analysis

$z = \dfrac{x - \mu}{\sigma}$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between −1 and 1
- approximately 95% of scores have z-scores between −2 and 2
- approximately 99.7% of scores have z-scores between −3 and 3

$E(X) = \mu$

$\operatorname{Var}(X) = E\left[(X - \mu)^2\right] = E\left(X^2\right) - \mu^2$

Probability

$P(A \cap B) = P(A)P(B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A|B) = \dfrac{P(A \cap B)}{P(B)}, \ P(B) \neq 0$

Continuous random variables

$P(X \leq x) = \displaystyle\int_a^x f(x)\,dx$

$P(a < X < b) = \displaystyle\int_a^b f(x)\,dx$

Binomial distribution

$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$

$X \sim \operatorname{Bin}(n, p)$

$\Rightarrow \ P(X = x)$

$= \dbinom{n}{x} p^x (1 - p)^{n-x}, \ x = 0, 1, \ldots, n$

$E(X) = np$

$\operatorname{Var}(X) = np(1 - p)$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\dfrac{dy}{dx} = nf'(x)[f(x)]^{n-1}$
$y = uv$	$\dfrac{dy}{dx} = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$
$y = \dfrac{u}{v}$	$\dfrac{dy}{dx} = \dfrac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\dfrac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\dfrac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\dfrac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\dfrac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\dfrac{dy}{dx} = (\ln a)f'(x)a^{f(x)}$
$y = \log_a f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{(\ln a)f(x)}$
$y = \sin^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \cos^{-1} f(x)$	$\dfrac{dy}{dx} = -\dfrac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \tan^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{1+[f(x)]^2}$

Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x)\sin f(x)dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + c$$

$$\int f'(x)a^{f(x)}dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

$$\int_a^b f(x)dx$$

$$\approx \frac{b-a}{2n}\left\{f(a) + f(b) + 2\left[f(x_1) + \cdots + f(x_{n-1})\right]\right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$$^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!\,(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$\left|\underset{\sim}{u}\right| = \left|x\underset{\sim}{i} + y\underset{\sim}{j}\right| = \sqrt{x^2 + y^2}$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = \left|\underset{\sim}{u}\right|\left|\underset{\sim}{v}\right|\cos\theta = x_1x_2 + y_1y_2,$$

where $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$

and $\underset{\sim}{v} = x_2\underset{\sim}{i} + y_2\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos\theta + i\sin\theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} \left[r(\cos\theta + i\sin\theta)\right]^n &= r^n(\cos n\theta + i\sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$x = a\cos(nt + \alpha) + c$$

$$x = a\sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$